RO Antiscalant Projection Engine

Technical review & improvement comments for the Chembond Flux development team

Prepared for: Rahil (tool author) → Chembond Chemicals web development team
Date: 7 November 2026 · **Updated 7 November 2026: fixes delivered in projection_engine.py v2.0.0**
Website tested: staging-console.chembondflux.com/projection/ (Admin account)
Source reviewed: GitHub repository **rahils20/Reliance-MED-Performance** (branch head 030a52a): projection_engine.py, streamlit_app.py, med_suite.py, calculator_tab.py, run_app.py, requirements.txt, .github/workflows/build-exe.yml

1. Scope and status

This review covers the original Python/Streamlit tool that Chembond Flux is based on. The calculation engine was read line by line, and its key functions were run on benchmark waters to confirm each finding with real numbers rather than by reading the code alone. Everything marked “verified” below was reproduced by running the code.

The Chembond Flux staging website was also tested (Section 2): five benchmark waters were entered through the New Analysis form, and the results, recommendations and finalised reports (web, PDF, Word) were compared with the original engine. Flux reproduces the original calculations exactly, so the engine defects in Sections 3–4 apply to the website as well, alongside the website-specific findings W1–W10.

Status: the engine findings are now fixed in the updated projection_engine.py v2.0.0 shared with this document (see the Update section and the status line under each finding). Two items still need Chembond data (product limits, k-factor calibration), and the Flux website must adopt v2.0.0.

Headline findings (original review)

- **Sulfate and fluoride saturation is overstated by about 3×** at typical RO concentrate salinity. The activity correction applied to the solubility products uses the wrong charge factor (verified: about ×3 at I ≈ 0.12 M). $CaSO_4$, $BaSO_4$, $SrSO_4$ and CaF_2 ratios, and therefore product selection and dose, are biased.
- **Zero calcium or zero bicarbonate switches off every scaling check.** A softened feed (Ca = 0) containing barium, strontium and sulfate reports zero risk for all salts, then tells the user that no catalogue product works (verified, case E). This matters for HERO-type plants, which run on softened feed.
- **Undersaturated calcium carbonate is treated as a failure.** The selector requires the treated LSI to be between −0.5 and +0.5. If the concentrate LSI is below −0.5 (soft or acidified water), every product fails and the UI shows “Standard Catalog Limits Exceeded – custom formulation required” (verified, case D).
- **The antiscalant model is an uncalibrated heuristic with no upper limit.** It lowers the thermodynamic SI itself. In one test, 7 ppm of a homopolymer-only product “controlled” gypsum at 26× saturation. No real antiscalant can do that, and supplier limits are typically about 2–4×.
- **The customer-facing report makes claims the model does not support.** The executive summary always says “severe scaling risk… permanent membrane damage is imminent”, even for benign water. The CIP-frequency and ROI figures are hard-coded, and the $CaCO_3$ row of the report table always prints 1.000.
- **Flux exposes Chembond's formulations:** every results page embeds the active-ingredient composition of all 10 Kem Watreat products and all k-factors as JSON in the page source (W2). The original code also hard-codes an admin override code and a master password.
- **Flux website:** the calculations match the original engine exactly (cases A–E), but the server accepts negative concentrations, 150 °C and pH 15 (W4). A 30%-imbalanced analysis is shown as “Balance Error 0.00% (OK)” (W5), and report links open without login for 7 days (W6).

Severity scale

Critical	Wrong answer or misleading customer output in common cases; fix before external use.
High	Material accuracy error or failure in a realistic scenario; data loss; security exposure.
Medium	Limited accuracy impact, edge-case defect, or misleading labelling.
Low	Code quality, maintainability or cosmetic issue.

Update: corrected engine delivered (projection_engine.py v2.0.0)

The changes marked “FIXED” in this document are made in the updated projection_engine.py (version 2.0.0) shared with this document. The same file is committed to the GitHub repository rahils20/Reliance-MED-Performance on branch claude/sleepy-pascal-ms92ho, together with a regression test suite (tests/test_projection_engine.py) and the benchmark script (docs/review/benchmark_cases.py). As agreed with the tool author, only the projection engine was changed. streamlit_app.py, med_suite.py, run_app.py and the build workflow are untouched, and their findings are marked “NOT CHANGED”. Every finding in Sections 2–6 now carries a status line.

How to use the file

- **As a standalone app:** `streamlit run projection_engine.py`. The interface is unchanged for streamlit_app.py, which still calls `UtilityProjectionEngine().render_engine()`.
- **From a backend (recommended for Flux):** `from projection_engine import run_projection`. It returns every result (indices, streams, acid dose, viable and rejected products with reasons, warnings) as a dictionary, and needs only the Python standard library. `build_docx_report()` produces the Word report.
- **Tests:** `python -m pytest tests/` (19 tests, all passing). The UI was also exercised end to end (standalone and through streamlit_app.py) with Streamlit's AppTest and in a real browser, with no exceptions.

Independent validation against PHREEQC

Concentrate saturation indices ($\log_{10}$ IAP/Ksp) from v2.0.0 compared with PHREEQC 3 (phreeqc.dat) on identical compositions. v1 values are shown for contrast.

Case / salt	v1 engine	v2.0.0	PHREEQC	v2 − PHREEQC
A $CaSO_4$	+0.226	−0.501	−0.490	−0.011
A $BaSO_4$	+1.554	+0.699	+0.675	+0.024
A CaF_2	+0.375	−0.069	−0.096	+0.027
A amorphous SiO_2	−0.319	−0.304	−0.291	−0.013
D $CaSO_4$	+0.835	−0.015	−0.002	−0.013
D $BaSO_4$	+2.339	+1.149	+1.151	−0.002
E $BaSO_4$ (Ca = 0)	0 (not evaluated)	+1.470	+1.460	+0.010
A $CaCO_3$ precipitable mass	×0.05 factor	417 mg/L	417 mg/L	0%
B $CaCO_3$ precipitable mass	×0.05 factor	1064 mg/L	1047 mg/L	+1.6%
E $BaSO_4$ precipitable mass	–	0.971 mg/L	0.977 mg/L	−0.6%

Agreement is within ±0.05 log units for brackish concentrates (ionic strength up to about 0.2 M). Above 0.5 M (seawater) the Davies model drifts by 0.1–0.25 and the engine now shows a warning. The LSI is kept as the industry-standard index; it differs from PHREEQC's calcite SI by design (total Ca and alkalinity, no ion pairing).

Benchmark cases after the fix (same inputs as Section 2)

Case	v1 / Flux today	v2.0.0 concentrate	v2.0.0 recommendation (ppm)
A	$CaSO_4$ 1.68 “High”, $BaSO_4$ 35.8, CaF_2 2.37; 6 products, R 4001 7.5 listed first	LSI 2.07; $CaSO_4$ 0.32, $BaSO_4$ 5.0, CaF_2 0.85	R 6196 4.0 · R 6863 5.0 · R 3687 5.0 · R 170 5.5 · R 428 I 7.0 · R 4001 7.5 (R 824, R 246 exceed the LSI limit)
B	$BaSO_4$ 156, $CaSO_4$ 4.9; R 170 7.0	LSI 2.48 after 21.6 mg/L H_2SO_4; $BaSO_4$ 15.4, SiO_2 1.13	R 6863 7.5 only. Silica above 1.0 rejects the products without a silica dispersant (placeholder limits)
C	All at 1.0; report says “severe scaling risk”	LSI −0.22; nothing supersaturated	All 10 at 1.0, with summary text “scaling potential is low”
D	“Catalog Limits Exceeded” (B1)	S&DSI governs; $CaSO_4$ 0.97, $BaSO_4$ 14.1	All 10 at 1.0, plus a −30% ion-balance warning
E	All indices 0, “Catalog Limits Exceeded” (B2)	LSI n/a; $BaSO_4$ 29.5, $SrSO_4$ 0.81	All 10 at 1.0, plus a −26% ion-balance warning

F	Saved and calculated	Rejected with 6 validation errors	–

Actions for Chembond before customer use of v2.0.0

- **Replace the placeholder product limits** in PRODUCT_LIMITS (max LSI, $CaSO_4$, $BaSO_4$, $SrSO_4$, CaF_2, silica per product) with values from Chembond induction-time/jar tests. The recommendations in case B depend directly on them.
- **Confirm or calibrate the k-factors** (K_FACTORS), which drive the required dose; they are unchanged from v1.
- **Enter the product SG and price** (PRODUCT_SG, PRODUCT_PRICE_PER_KG) to get correct L/h figures and cost-based ranking.
- **Set the admin code** for custom blends in .streamlit/secrets.toml as projection_admin_code. Without it, custom synthesis is disabled. Rotate the old code, which was public in the source.

Other changes in v2.0.0

- **New issues found and fixed during the update.** (B10) A manual “adjusted pH” above the raw pH was used for the concentrate pH although no caustic is dosed; it is now rejected with a warning. (B11) Ion inputs and the balance checkbox reset whenever another input changed, because their widget identity depended on their value; they now use stable keys.
- **Default water and labels.** The default water is now balanced (Cl 587.3 mg/L) and its CO_2 is consistent with pH 7.5 (12.8 mg/L). Tab 3 is renamed “Formulation Engine”, because the selection is rule-based rather than AI.
- **Note for the whole app (outside this update):** Streamlit has deprecated use_container_width, which every module uses. Pin the Streamlit version in requirements.txt until all modules move to width='stretch'.

2. Chembond Flux website (staging) – review results

Tested on 7 Nov 2026 at staging-console.chembondflux.com/projection/ with the rahil.shah account (Admin role), using headless Chromium at desktop and iPhone-13 sizes. Each benchmark case was entered through the New Analysis form, then the results, the AI Formulation tab, the finalised report page and the downloaded PDF and Word reports were checked.

2.1 Calculation parity with the original engine

Result: Flux reproduces the original engine exactly. Every value below matched to the last displayed decimal: acid dose, treated pH, concentrate LSI/SDSI, ionic strength, all saturation ratios, and the product list, order and doses. The port itself is faithful, which also means **every engine defect in Sections 3–4 is live on the website** (see W1). Common inputs unless stated: Ca 150, Mg 50, Na 300, K 10, NH_4 0, Ba 0.05, Sr 1.2, Fe 0.02, Al 0.01, HCO_3 250, Cl 400, SO_4 200, F 0.5, NO_3 5, PO_4 0, CO_3 0, SiO_2 15, CO_2 5 mg/L; 99% rejection; BWRO; feed 10 m³/h.

Case	Inputs (changes from default)	Treated pH / H_2SO_4 ppm	Conc. pH / LSI	I (M)	$CaSO_4$ / $BaSO_4$ / $SrSO_4$ / CaF_2 ratio	Recommendations (ppm)	Flux
A	Default water, pH 7.5, 25 °C, 75%, Na/Cl balance on (existing “test” analysis)	7.50 / 0.00	8.102 / 2.065	0.1313	1.684 / 35.84 / 0.457 / 2.372	R 4001 7.5 · R 428 I 7.0 · R 6196 4.0 · R 6863 5.0 · R 170 5.5 · R 3687 5.0	✓ identical
B	Ca 200, HCO_3 400, SO_4 300, Ba 0.1, Sr 2.0, SiO_2 30; pH 7.8, 30 °C, 80%; balance on	7.30 / 21.56	7.999 / 2.472	0.1893	4.888 / 156.04 / 1.657 / 5.757	R 170 7.0 only	✓ identical
C	Soft: Ca 10, Mg 3, HCO_3 30, SO_4 20, Cl 60, Na 40, Ba/Sr/F 0; pH 7.0; balance off	7.00 / 0.00	7.602 / −0.207	0.0132	0.022 / 0 / 0 / 0	All 10 at 1.0 ppm	✓ identical
D	Ca 100, HCO_3 8, SO_4 1500, Na 300, Cl 200, Mg 20; pH 6.0, manual acid; balance on	6.00 / 0.00	6.602 / −1.176	0.2242	6.839 / 218.33 / 2.783 / 1.425	None → “Catalog Limits Exceeded” (B1)	✓ identical
E	Softened: Ca 0, Mg 0.5, Ba 0.15, Sr 3.0, HCO_3 200, SO_4 400; balance on	7.50 / 0.00	8.102 / −5.000	0.0000	0 / 0 / 0 / 0	None → “Catalog Limits Exceeded” (B2)	✓ identical

2.2 Website findings

ID	Severity	Finding, location & evidence	Recommendation & status in v2.0.0
W1	**Critical**	**Every engine defect is live on Flux.** Confirmed on staging: sulfate/fluoride ratios about 3× high (M1; e.g. case A $CaSO_4$ 1.684 flagged “High”). Cases D and E produce “Standard Catalog Limits Exceeded. No standard product can safely maintain this water profile below 8.0 ppm” (B1, B2). A softened feed with Ba/Sr/SO_4 shows ionic strength 0.0000 and every salt “Okay”. The antiscalant model is unchanged (M11, M12).	Fix the engine first (Sections 3–4), then re-run cases A–E as automated regression tests in the Flux codebase. **FLUX WEB TEAM:** Engine defects are fixed in v2.0.0. Flux must port v2.0.0 (or call run_projection() server-side).
W2	**Critical**	**Proprietary formulation data is sent to the browser.** Every results page embeds two JSON blocks, `px-formulations-data` (active % of PBTC, DETMPA, HEDP, ATMP, polymers, SMBS for all 10 Kem Watreat products) and `px-k-factors-data` (all k-factors), to drive the Project X grid in JavaScript. Anyone who can open a results page can read them with View Source.	Compute the Project X grid on the server and send only the resulting percentages. Restrict formulation data to an authorised role, and check which user roles (including customer logins, if any) can reach /projection/. **FLUX WEB TEAM:** Web team action. The v2 engine needs no product data in the browser.

ID	Severity	Finding, location & evidence	Recommendation & status in v2.0.0
W3	**Critical**	**The customer report (web page, PDF and Word) inherits the report defects and adds new ones.** Verified on the case B PDF: the hard-coded “severe scaling risk... permanent membrane damage is imminent... completely suppress” summary (B3), and the hard-coded CIP/ROI block (B4). In the table, $CaCO_3$ goes from 1.000 to **2.911**, so the treated case looks about 3× worse (B5). $CaSiO_3$/$MgSiO_3$ go from 0.000 to 1.000, and any salt reduced below saturation is printed as 1.000 (“at saturation”). The report omits the water analysis, feed flow, temperature, CF, chemical flow (L/h), assumptions, model version, prepared-by and a disclaimer. Improvements over the original: the site and the acid dose and treated pH are now included.	Generate the summary from the results, remove the ROI block, fix the treated-value mapping (show the actual SI, not a clipped ratio), and add an inputs annex plus an assumptions, version and disclaimer block. **FLUX WEB TEAM:** Report fixed in v2.0.0 (build_docx_report). Flux must update its web, PDF and Word templates to match.
W4	**High**	**No server-side input validation.** An analysis was saved and calculated with Ca −50, HCO_3 −10, SO_4 −5 mg/L, feed flow 0 m³/h, 150 °C, feed pH 15 and permeate pH −2. The results show negative concentrations, TDS 721.78 mg/L and a “Catalog Limits Exceeded” verdict. The HTML inputs have no min/max for ions, flow, temperature or pH.	Add Django model/form validators: concentrations ≥ 0, flow > 0, temperature 5–45 °C, pH 4–10, Ca > 0 or an explicit softened-feed path. Also mirror them as HTML min/max. **FLUX WEB TEAM:** Reference rules are in validate_inputs(); Flux must enforce them server-side.
W5	**High**	**A bad lab analysis looks clean after auto-balance.** Case D was entered 30% out of balance, a strong sign of a lab or transcription error, but the results page shows a green “Balance Error: 0.00% (OK)”. The “Raw Feed” column shows the adjusted Na/Cl (case A: Cl 587.30 vs 400 entered). The entered values are kept in the edit form, so the data exists but is not displayed.	Show the entered analysis, the original imbalance and the Na/Cl added as separate items. Warn (or block finalising) above about 10% imbalance, and print the adjustment in the report. **FLUX WEB TEAM:** The v2 engine shows the entered imbalance and the adjustment applied, and prints both in the report. Flux must mirror this.
W6	**Medium**	**Report download links work without login for about 7 days.** “Download PDF/Word Report” are S3 pre-signed URLs with X-Amz-Expires = 604700 s. A forwarded link opens the customer proposal without authentication.	Serve reports through an authenticated Django view that issues a short-lived (minutes) pre-signed URL on each click. **FLUX WEB TEAM:** Web team action (short-lived links via an authenticated view).
W7	**Medium**	**Inconsistent report timestamps.** For the same finalisation, the PDF says “Date Generated: 2026-11-07 22:26” and the Word file says “16:56”. That looks like local India time vs UTC, and neither shows a time zone.	Use one timezone-aware timestamp (USE_TZ, localised to the site's time zone) for both formats, and print the zone. **FLUX WEB TEAM:** v2 prints one labelled IST timestamp. Flux must do the same in both formats.
W8	**Medium**	**Charts mislead.** (a) Before/after scaling potential uses a linear % axis: $BaSO_4$ at about 15,500% flattens every other salt and the “After” bars are invisible (web and PDF). (b) The performance-curve caption promises a green Safe Zone line and a red finalised-dose line, but neither is drawn on the web chart. (c) Fe is plotted as a flat line at about 2.0 on the “Effective SI” axis, which looks like an uncontrolled scale (it is mg/L ÷ 0.05). (d) The baseline chart has no y-axis title or units.	Use a log scale or plot SI/ratio directly, draw the promised reference lines, remove Fe/Al from the SI chart (show them as a separate pre-treatment advisory), and label the axes. **FLUX WEB TEAM:** v2 charts use a log SI scale, draw the reference lines and exclude Fe. Flux must mirror this.
W9	**Low**	**Labels and UX.** “Mineral Passage: 0.0100” is a fraction (the original showed 1.0%). The “Slightly Soluble Salts ... — Treated Concentrate” heading sits over a table showing all five streams. The list-page filter drop-downs have no labels (they show “---------”). “Feed Temperature (C)” is missing the degree symbol. Finalise-dose and custom-dose inputs accept up to 20 ppm, while the search stops at 8 ppm. Custom-blend percentages are not checked to total ≤ 100%.	Tidy the labels and units, add filter placeholders (Site / Status / Product), align the dose limits, and validate the blend total. **FLUX WEB TEAM:** Engine-side labels fixed in v2 (salt passage %, °C, dose limits, blend total). The list-page filter labels are a Flux change.

ID	Severity	Finding, location & evidence	Recommendation & status in v2.0.0
W10	**Low**	**What works well:** login is enforced (anonymous requests redirect to the login page). Forms carry CSRF tokens. The page has no horizontal scroll at phone width. Edit Inputs keeps the as-entered values. The acid dose and treated pH are shown prominently. Analyses are linked to sites and have Draft/Finalised status. The hard-coded “KEMPRO2026” prompt has been replaced: the custom-blend builder appeared directly for the Admin role.	Confirm that non-admin roles cannot open the custom-blend builder or finalise analyses. Add an audit trail of who finalised which proposal. **FLUX WEB TEAM:** No action needed.

Test records created on staging (please delete after review): five analyses under Temp Site 1 named “ZZ Claude review – B hard 80pct / C soft / D low-alk manual pH6 / E softened Ca0 / F invalid inputs”. Case B was finalised with Kem Watreat R 170 at 7.0 ppm, which generated a PDF and a Word report on S3. Not yet checked: non-admin role permissions, Firefox/Safari, and performance under load.

3. Calculation engine – accuracy findings (projection_engine.py)

What is correct: the carbonate constants (Harned–Scholes pK_2 = 10.330, Plummer–Busenberg calcite pK_s = 8.480, Harned–Davis pK_1 = 6.351 at 25 °C), the Langelier pH_s expression with Davies activity coefficients, the H_2SO_4 stoichiometry (98.08 per 2×61.02, ÷0.98) and the chemical feed-rate arithmetic were all checked and are sound. The issues below concern the other salts, the antiscalant model and the system mass balance.

3.1 Thermodynamics & water chemistry

ID	Severity	Finding, location & evidence	Recommendation & status in v2.0.0
M1	**Critical**	**Wrong charge factor in the Ksp activity correction.** get_pKsp_apparent() (lines 136–152) uses z = 4 for $CaSO_4$/$BaSO_4$/$SrSO_4$/$CaSiO_3$ and z = 2 for CaF_2. Debye–Hückel requires Σz_i^2 = 4+4 = 8 for a 2:2 salt and 4+2×1 = 6 for CaF_2. Verified: at the default concentrate (I ≈ 0.12–0.13 M) the apparent Ksp is understated about 3× (×2.99–3.09), so every sulfate/fluoride ratio is about 3× too high. Example: case A $CaSO_4$ shows 1.68 ("High") where the corrected value is about 0.55 (undersaturated).	Use the same Davies γ already computed for LSI: $Ksp' = Ksp / (\gamma_{cat}\cdot\gamma_{an}^{n})$. Use one activity model throughout. Add unit tests against PHREEQC for 3–5 reference waters. **FIXED in projection_engine.py v2.0.0 –** Ksp now corrected with the same Davies activity coefficients as the LSI; matches PHREEQC within ±0.05 log units.
M2	**High**	**No ion pairing.** Free-ion activities ignore the $CaSO_4$°, $MgSO_4$°, $NaSO_4^-$, $CaHCO_3^+$ and $CaCO_3$° complexes. In sulfate-rich or high-TDS concentrates this overstates sulfate scaling (by tens of percent at high SO_4) and also biases LSI.	Use a proper speciation engine: PHREEQC (phreeqpy/IPhreeqc, with phreeqc.dat or pitzer.dat for SWRO) on the server. At minimum, add $CaSO_4$° and $MgSO_4$° pairing. **FIXED in projection_engine.py v2.0.0 –** Ion pairing added (CaSO4°, MgSO4°, NaSO4−, KSO4−, SrSO4°, BaSO4°, CaHCO3+, MgHCO3+, MgF+) with phreeqc.dat constants.
M3	**High**	**No ionic strength or other index when Ca = 0 or HCO_3 = 0.** Lines 108–118 return LSI = −5 and zero for every sulfate, fluoride and silica index. Verified (case E): softened feed with Ba 0.15, Sr 3.0, SO_4 400 returns I = 0 and $BaSO_4$ = $SrSO_4$ = 0. The same happens with fully dealkalised feed.	Only skip the LSI/SDSI calculation when Ca or HCO_3 is zero; compute all other salts normally. (Also see B2 for the selector impact.) **FIXED in projection_engine.py v2.0.0 –** Ca = 0 or HCO3 = 0 now only switches off LSI/S&DSI; all other salts and the ionic strength are still calculated.
M4	**Medium**	**Silica limit has no pH or temperature basis beyond a linear fit.** The limit is fixed at 125 mg/L at ≤ 25 °C (lines 166–171) with no pH correction. Verified: the ratio is identical at concentrate pH 6.0, 7.0, 8.0 and 8.5. Amorphous silica solubility rises sharply above about pH 8.5, and vendor programs apply a pH correction factor. The same limit is coded four times (lines 166, 299, 947, 1145).	Implement the standard temperature + pH correction for amorphous silica (with Mg/Al co-precipitation warnings). Define the limit once in a shared function. **FIXED in projection_engine.py v2.0.0 –** Silica solubility from Fournier & Rowe (temperature) with a pH correction (pKa 9.84), defined once.
M5	**Medium**	**Metal silicate model is not referenced.** The SiO_3^{2-} fraction uses pKa_2 = 11.8 (literature for H_4SiO_4 is about 13). pKsp($CaSiO_3$) = 8.0, and Mg = 1.5× / Fe = 0.05× of the Ca value, have no source. A hard cut-off at pH 8.0 makes results jump. The report prints $CaSiO_3$/$MgSiO_3$ rows as if they were validated.	Remove these indices from customer outputs until validated, or replace them with an established magnesium-silicate guideline (Mg × SiO_2 vs pH). **FIXED in projection_engine.py v2.0.0 –** Metal-silicate indices removed; replaced by a qualitative warning above concentrate pH 8.5.
M6	**Medium**	**The carbonate system is over-specified.** Feed pH, HCO_3 and CO_2 are entered independently and never reconciled; the CO_2 input does not affect any result. CO_2 released by acid (≈0.72 mg per mg HCO_3 destroyed) is not added to the treated feed. Concentrate pH = feed pH + $\log_{10}$CF (lines 496, 523) assumes 100% CO_2 passage and ignores CO_3^{2-}. This approximation is acceptable, but it is undocumented.	Calculate CO_2 from pH and alkalinity, and warn when the entered CO_2 disagrees. Carry the acid-generated CO_2. Document the concentrate-pH method (e.g. ASTM D4582) in the UI and report. **FIXED in projection_engine.py v2.0.0 –** CO2 calculated from pH and alkalinity (acid-generated CO2 included); the entered CO2 is cross-checked with a warning.

ID	Severity	Finding, location & evidence	Recommendation & status in v2.0.0
M7	**Medium**	**Mass balance is not consistent.** The concentrate uses $C_f \times CF$ and ignores salt passage (exact value: $C_f(1-Y \cdot SP)/(1-Y)$; ≈1% difference at 99% rejection). Permeate is computed from the untreated feed rather than the acid-treated feed, and from the feed concentration rather than the average feed–concentrate concentration. **Membrane type has no effect on any calculation**; it only changes the recommended-recovery caption.	Use the exact mass balance with the treated feed. Either remove the membrane-type selector or tie it to rejection, recovery limits and the SWRO activity model. **FIXED in projection_engine.py v2.0.0 –** Exact mass balance with salt passage, from the treated feed. Membrane type now sets rejection/recovery defaults and an SWRO warning.
M8	**Medium**	**Constants fixed at 25 °C.** Sulfate and fluoride pKsp values ignore temperature, although the input allows 1–50 °C ($BaSO_4$ and CaF_2 solubility change noticeably). pKsp sources are not cited ($SrSO_4$ 6.49 vs ≈6.63 and CaF_2 10.40 vs ≈10.5–10.6 in common databases).	Add temperature-dependent log K (van 't Hoff or analytic expressions from phreeqc.dat). Cite the source of each constant in the code and the report. **FIXED in projection_engine.py v2.0.0 –** log K from phreeqc.dat analytical expressions (van 't Hoff where none exists), with sources cited in one table.
M9	**Low**	**SDSI is not Stiff–Davis.** It is the LSI with a different activity term (line 133), not the empirical Stiff–Davis K. The selector also forces both LSI and SDSI into ±0.5, although SDSI is meant for TDS above about 10,000 mg/L and LSI for brackish water.	Pick the index by TDS (LSI for brackish, S&DSI for SWRO), or show both with clear applicability notes. **FIXED in projection_engine.py v2.0.0 –** S&DSI governs above 10,000 mg/L concentrate TDS, LSI below. The S&DSI is still an activity-corrected approximation.
M10	**Low**	**TDS and balance details.** Calculated TDS adds dissolved CO_2 (a gas) to TDS. The Fe equivalent weight assumes Fe^{2+}. "Fe/Al SI" is just mg/L ÷ 0.05, a feed guideline that is then applied to the concentrate and plotted on an SI axis. The app's own default water is 11.9% out of balance, so auto-balance runs on first load.	Exclude CO_2 from TDS. Label Fe/Al as a feed-guideline ratio and evaluate it on the feed. Ship a balanced default water. **FIXED in projection_engine.py v2.0.0 –** TDS excludes CO2; Fe labelled as Fe2+; Fe/Al shown as a feed pre-treatment advisory only; the default water is balanced.

3.2 Antiscalant performance model & product selection

ID	Severity	Finding, location & evidence	Recommendation & status in v2.0.0
M11	**Critical**	**The antiscalant model changes the thermodynamic SI and is not calibrated.** calculate_effective_scaling() (lines 241–279) sets $SI_{eff} = SI \cdot \exp(-\Sigma \text{ dose} \cdot \text{active\%} \cdot k / \sqrt{SI})$. Antiscalants do not change saturation. They extend induction time, which raises the saturation the system can tolerate. The k-factors (lines 43–53) have no data source. **There is no upper limit:** verified that R 824 (homopolymer only) at 7 ppm "controls" $CaSO_4$ at 26× saturation (about 9× even after the M1 fix). Supplier limits are typically about 2–4× for gypsum and LSI about 1.8–2.8.	Reformulate the model as product maximum allowable limits vs dose (LSI_{max}, $CaSO_4$×, $BaSO_4$×, $SrSO_4$×, CaF_2×, SiO_2 mg/L), derived from Chembond induction-time/jar tests and field history, and compare them with the actual concentrate values. calculator_tab.py already produces such a limits matrix: connect it, then validate it. Keep reporting the real SI. **PARTLY FIXED in projection_engine.py v2.0.0 –** Per-product maximum-limit table added and enforced (PRODUCT_LIMITS). **The values are placeholders and the k-factor dose model is unchanged; both need Chembond test data.**
M12	**High**	**$BaSO_4$/$SrSO_4$ can never fail the selection.** A salt only fails if its "excess mass" is above 0.5 ppm (line 334). The maximum possible $BaSO_4$ in the default concentrate is 0.34 mg/L, so barium can never govern, even at 36× saturation (case A) or 156× (case B). Barium sulfate is one of the most damaging and least reversible RO scales.	Judge sparingly soluble salts (Ba, Sr, CaF_2) on saturation against the product limit, not on mass. A mass threshold is only meaningful for $CaCO_3$/$CaSO_4$. **FIXED in projection_engine.py v2.0.0 –** BaSO4/SrSO4/CaF2 judged on saturation against the product limit; the 0.5 ppm mass escape is removed.

ID	Severity	Finding, location & evidence	Recommendation & status in v2.0.0
M13	**Medium**	**The excess-mass calculation is arbitrary.** $CaCO_3$ mass is multiplied by 0.05 with no stated basis (lines 293, 942, 1139). The (1−1/ratio) precipitable fraction is only approximate. For silica, the excess above the limit is multiplied by (1−1/ratio) again, which double-counts. The CaF_2 exemption up to SI 1.5 (line 335) is undocumented. This function is copy-pasted three times.	Calculate precipitable mass by solving the equilibrium (mass-balanced IAP = Ksp′) or with PHREEQC. Keep one shared implementation. **FIXED in projection_engine.py v2.0.0 –** Precipitable mass solved from equilibrium (closed-system CaCO3), one shared function; matches PHREEQC within 3%.
M14	**Medium**	**The “economic” order ignores dose and price.** economic_sort_order (line 283) is a fixed list. R 4001 is ranked first although it needs 7.5 ppm vs 4.0 ppm for R 6196 (case A).	Rank by cost per m³ = dose × price (and by product availability). Show the cost per m³ and per year in the table. **FIXED in projection_engine.py v2.0.0 –** Viable products ranked by required dose, or by cost per m³ once prices are entered (PRODUCT_PRICE_PER_KG).
M15	**Medium**	**Auto-acid only looks at LSI.** Lines 486–505 target concentrate LSI ≤ 2.5, a fixed value that is not tied to the chosen antiscalant. H_2SO_4 adds sulfate, which raises $CaSO_4$/$BaSO_4$. Verified: acid to pH 6 raised the $CaSO_4$ ratio from 1.74 to 2.84. HCl is not offered. If LSI ≤ 2.5 is not reached by pH 4, no acid is applied and no warning is shown.	Optimise acid against all the limits together. Offer HCl. Warn when the target is not reachable. Tie the LSI target to the selected product's limit. **FIXED in projection_engine.py v2.0.0 –** Choice of H2SO4 or HCl, adjustable index target, pH floor 5.0, and warnings when the target is unreachable or CaSO4 rises.
M16	**Low**	**Dose basis is ambiguous.** The Project X slider is labelled “Active Product Dose” but is multiplied by active % (line 264), so it is really the product dose. The UI never says whether the dose is on a feed or concentrate basis.	Label it “Product dose, mg/L on feed flow”, and optionally show the active dose and the concentrate-side concentration. **FIXED in projection_engine.py v2.0.0 –** Labelled as product dose in mg/L on feed flow throughout.

4. Engine logic defects & customer report (projection_engine.py)

ID	Severity	Finding, location & evidence	Recommendation & status in v2.0.0
B1	**Critical**	**Undersaturated LSI is treated as a failure.** Lines 323–324 require $-0.5 \le LSI_{eff} \le 0.5$. An antiscalant cannot raise LSI, so a concentrate with LSI < −0.5 fails for every product. The user then sees “Standard Catalog Limits Exceeded... Custom formulation required” and an admin-code prompt. Verified: case D (LSI −1.18, $CaSO_4$ 6.8×) gets zero products, while the same engine accepts 26× $CaSO_4$ when LSI is positive.	Apply an upper limit only (LSI ≤ product limit). Treat negative LSI as “no $CaCO_3$ risk” and, if very negative, show a corrosivity note. **FIXED in projection_engine.py v2.0.0 –** Upper limit only; negative LSI is treated as no CaCO3 risk (case D now gets products).
B2	**High**	**A softened or dealkalised feed causes a false “no product works”.** Because of M3, all indices are zero and LSI = −5, so B1 rejects every product (case E). A HERO plant (softener + HRU upstream) cannot be simulated.	Fix M3 and B1, then add regression test case E. **FIXED in projection_engine.py v2.0.0 –** Softened feeds now simulate correctly (case E: BaSO4 29.5×, products found).
B3	**Critical**	**The executive summary is hard-coded.** Lines 907–912 always state “severe scaling risk... rapid loss of flux and permanent membrane damage is imminent” and that the product will “completely suppress” the salts. This is printed even for case C (LSI −0.21, every product passes at 1 ppm). It is a commercial and legal liability in a customer proposal.	Generate the text from the results (risk class by salt). Use measured language (“projected”, “based on the supplied analysis”). Add assumptions and a disclaimer. **FIXED in projection_engine.py v2.0.0 –** Executive summary generated from the results, with measured wording, site-monitoring advice and a disclaimer.
B4	**High**	**CIP and ROI figures are invented.** In lines 1039–1051 the “b_max_mass_ppm” variable adds up saturation ratios, not masses. Using 10^{LSI} means any LSI above about 1.7 gives “CIP every 1–2 weeks”. The treated case is hard-coded to “CIP every 3–6 months, membrane life optimal, energy stable” whatever the dose or water.	Remove this section, or base it on site history or a documented fouling model. Never show it as a projection without a basis. **FIXED in projection_engine.py v2.0.0 –** CIP/ROI block removed from the screen and the report.

ID	Severity	Finding, location & evidence	Recommendation & status in v2.0.0
B5	**High**	**The $CaCO_3$ row of the report table is always 1.000.** Line 968 looks up “Ratio_CaCO3”, which does not exist, so the default of 1.0 is printed as the baseline.	Show the LSI (or 10^{LSI}) for $CaCO_3$ consistently, or drop the duplicate row. **FIXED in projection_engine.py v2.0.0 –** Report table shows concentrate value vs product limit and status; no clipped 1.000 rows.
B6	**High**	**Report content is incomplete.** The .docx omits the water analysis (raw and balanced), temperature, pH, flow, CF, acid dose, chemical flow (L/h), the product limits used, and the software/model version. Charts are silently omitted when kaleido cannot render (kaleido ≥ 1.0 needs Chrome). Temporary PNG files are never deleted (delete=False, line 1014).	Add an inputs annex and an assumptions/version block. Use BytesIO for images, or render charts with matplotlib (already a dependency). **FIXED in projection_engine.py v2.0.0 –** Report adds specification, inputs annex, assumptions, engine version and labelled timestamp; charts via matplotlib in memory.
B7	**Medium**	**Errors are hidden.** calculate_scaling_indices() catches every exception and returns None (lines 238–239), and the UI then shows only “ensure Ca and HCO_3 > 0”. Results depend on `'treated_conc_data' in locals()` (lines 800, 890).	Validate inputs explicitly, show the actual error, and pass results as return values or a dataclass. **FIXED in projection_engine.py v2.0.0 –** validate_inputs() checks every input and errors are shown; no reliance on locals().
B8	**Medium**	**Custom blend issues.** Percentages are not checked to sum to ≤ 100% (lines 836–850). The success message is lost because st.rerun() runs straight after it. The custom blend stays in the session and pollutes later product lists.	Validate the blend total, persist the message across the rerun, and clear the blend on reset. **FIXED in projection_engine.py v2.0.0 –** Blend total must be 0–100%; the confirmation survives the rerun; a “Clear custom blend” button was added.
B9	**Low**	**Other UI issues.** The finalise-dose input allows up to 10 ppm, but the search stops at 8. The print CSS relies on auto-generated .st-emotion-cache class names that change between Streamlit versions. st.set_page_config() runs when the module is imported (line 10), before the hub's own call: older Streamlit raises an error, and newer versions depend on call order.	Align the limits, move the set_page_config call under __main__, and pin the Streamlit version. **FIXED in projection_engine.py v2.0.0 –** Dose limits aligned to 1–8 ppm; set_page_config only when run standalone; print CSS uses stable selectors.

5. Security, data integrity & RO monitoring hub (streamlit_app.py, med_suite.py)

ID	Severity	Finding, location & evidence	Recommendation & status in v2.0.0
S1	**Critical**	**Hard-coded secrets.** The admin override “KEMPRO2026” is in projection_engine.py line 826, and the master password “12345678” appears 3× in streamlit_app.py and 7× in med_suite.py. Anyone with repository access can read them. The chatbot even refers to the “Master Password”.	Move secrets to st.secrets or a secret manager, store only hashes, and use per-user authentication (SSO in Flux) with roles and an audit trail. Rotate both values now. **PARTLY FIXED in projection_engine.py v2.0.0 –** Admin override code removed from the engine; it is read from st.secrets[“projection_admin_code”]. The master password in streamlit_app.py and med_suite.py is unchanged (outside the agreed scope).
S2	**High**	**Proprietary data is exposed.** The product compositions (active % of PBTC, HEDP, ATMP, polymers) and k-factors are in plain code, and Flux sends them to the browser on every results page (W2).	Keep them server-side and encrypted at rest. Never ship them in a client bundle or API response. **FLUX WEB TEAM:** The engine keeps product data server-side in Python. Flux must stop embedding it in pages (see W2).

ID	Severity	Finding, location & evidence	Recommendation & status in v2.0.0
S3	High	**RO data and the RO model are not persisted.** load_database()/save_database() use Google Sheets only when target_file == LOCAL_DB_FILE (MED), lines 97 and 118. RO records and the RO RF/XGB model (joblib, line 1133) go to the ephemeral local disk, which Streamlit Cloud wipes on restart. When the model file is missing, the prediction becomes 0 (line 776), the residual % becomes 0, and the dashboard shows **"CLEAN"**, hiding fouling.	Persist RO data and models the same way as MED (or in the Flux database). If the model is missing, show "Model unavailable" instead of a status. **NOT CHANGED:** outside the agreed scope for this update (projection engine only).
S4	High	**Silent write failures.** Bare `except: pass` around the Sheets connection and writes (init_db_connection, save_database; 8 bare excepts in streamlit_app.py) means "Record saved" is shown even when only the local file was written.	Report the storage target and any failure to the user, and log exceptions. **NOT CHANGED:** outside the agreed scope for this update (projection engine only).
R1	High	**The RO "normalised twin" is not normalised.** Predicted permeate = f(Feed Flow, Feed TDS, Coagulant ppm, SMBS ppm) (lines 761–770). It has no temperature term (flux changes about 3%/°C), no feed pressure/NDP, no ΔP and no osmotic correction. Coagulant and SMBS have no physical link to permeate flow. With CIP thresholds at −4%/−5%, a few degrees of seasonal temperature change will trigger "Membrane CIP Required".	Implement ASTM D4516 normalisation: normalised permeate flow, normalised salt passage and normalised ΔP, against a post-CIP baseline. Use the industry triggers (NPF −10–15%, ΔP +15%, NSP +5–10%). Keep ML only as a supplement. **NOT CHANGED:** outside the agreed scope for this update (projection engine only).
R2	Medium	**Model accuracy is overstated.** R^2 is computed on the training data (lines 1092–1099), so RF/XGB will show close to 100%.	Use a time-based hold-out or cross-validation, and report MAE in m^3/h. **NOT CHANGED:** outside the agreed scope for this update (projection engine only).
R3	Medium	**Bulk upload invents data.** Missing cells are filled with 450 m^3/h, 2000 ppm TDS, TSS 8, SDI 2.5, pH 7.2 and so on (lines 1175–1215). Pass/fail is then graded on these values, and a blank permeate flow becomes 0, which raises a false fouling alarm. The date is parsed with dayfirst=True (line 1182), unlike the ISO-first parsing fixed elsewhere.	Leave missing values as missing (NaN) and exclude them from grading and modelling. Reuse the shared ISO-first date parser. **NOT CHANGED:** outside the agreed scope for this update (projection engine only).
R4	Medium	**Antiscalant is missing from the RO hub.** The RO Chemical Dosing tab covers only coagulant, flocculant and SMBS, and says that optimal dosing is "in development". The chatbot says the same, although the Projection Engine exists.	Link the RO hub to the Projection Engine: pull the latest feed analysis, show the recommended and actual antiscalant dose, and log the dose. **NOT CHANGED:** outside the agreed scope for this update (projection engine only).
R5	Low	**Smaller RO issues.** "Save Operational Record" appends duplicate dates (line 954). Remarks loaded from the database (ro_remarks) are not shown in the text box (ro_in_remarks). The chatbot keyword "ai" matches "explain", "maintain" and "train" (line 476). The permeate pH guarantee of 7.0–7.5 is unusual for RO permeate (typically 5.5–6.5 before correction): confirm the specification.	Upsert by date, bind a single remarks key, match chatbot keywords on word boundaries, and confirm the guarantee table with the plant. **NOT CHANGED:** outside the agreed scope for this update (projection engine only).
R6	Low	**MED chemical constants.** Antiscalant SG is 1.20 in MED (med_suite.py line 368) but 1.15 in the projection engine. AF_DENSITY = 0.02 is a dilution factor, not a density. LITRES_PER_LEVEL_UNIT = 23 is undocumented tank geometry.	Use a single product master table (SG, active %, price), and name the constants for what they are. **NOT CHANGED:** outside the agreed scope for this update (projection engine only).

6. Build, deployment & code quality

ID	Severity	Finding, location & evidence	Recommendation & status in v2.0.0
D1	High	**The desktop launcher exits immediately.** run_app.py starts Streamlit in a daemon thread, opens the browser, and the main thread then ends, which kills the server. Streamlit's CLI also registers signal handlers, which fail outside the main thread.	Run stcli.main() on the main thread, and open the browser from a threading.Timer. **NOT CHANGED:** outside the agreed scope for this update (projection engine only).

ID	Severity	Finding, location & evidence	Recommendation & status in v2.0.0
D2	High	**The executables are missing modules.** build-exe.yml bundles only streamlit_app.py. projection_engine.py, med_suite.py, calculator_tab.py, chembond_logo.png and .streamlit/config.toml are not added, so the build fails on import. plotly and kaleido are not installed in CI.	Add each file with --add-data (or use a .spec file), and add a smoke test that launches the built app in CI. **NOT CHANGED:** outside the agreed scope for this update (projection engine only).
D3	Medium	**Dependencies.** requirements.txt has no version pins. “plotly.express” is not the plotly package (it resolves to the deprecated plotly-express shim). streamlit is not listed. fpdf is unused. oauth2client is deprecated (use google-auth / gspread.service_account_from_dict).	Pin versions (pip-tools or uv lock), list plotly and streamlit explicitly, and remove unused packages. **NOT CHANGED:** outside the agreed scope for this update (projection engine only).
D4	Medium	**No tests or documentation.** README.md is empty, and there are no unit or regression tests for a calculation that drives customer proposals.	Add pytest regression tests for cases A–E plus PHREEQC cross-checks, and run them in CI. Write a README covering the method, assumptions, constants, sources and limitations. **PARTLY FIXED in projection_engine.py v2.0.0 –** Engine regression tests added (tests/test_projection_engine.py, 19 tests). README is still empty (outside the agreed scope).
D5	Low	**Structure.** projection_engine.py mixes UI and chemistry in one 900-line method, with functions copied three times. .devcontainer disables CORS and XSRF protection (acceptable only for local development).	Split it into a pure chemistry package (no Streamlit imports) that Flux can call through an API, plus a thin UI layer. That makes the web port and the tests share one engine. **PARTLY FIXED in projection_engine.py v2.0.0 –** Engine chemistry is now pure functions with a single run_projection() entry point Flux can call. The hub is unchanged.

7. Recommended order of work (remaining after v2.0.0)

Engine items M1–M16 (except the M11 data), B1–B10 and S1 (engine part) are fixed in projection_engine.py v2.0.0. Remaining work:

Priority	Items
1 – before any further customer use	Flux: port v2.0.0 or call run_projection() server-side (W1, W3, W4, W5); remove formulation data from pages (W2, S2); rotate the hub master password (S1, hub part)
2 – model credibility (needs Chembond data)	Replace the placeholder PRODUCT_LIMITS and calibrate the K_FACTORS from lab/field data (M11); enter product SG and prices (M14)
3 – web hardening & plant monitoring	W6, W7, W8; hub data persistence and silent failures (S3, S4); ASTM D4516 normalisation (R1); R2, R3, R4
4 – engineering hygiene	W9; D1–D3, D4 (README), D5 (hub); R5, R6; pin the Streamlit version (use_container_width deprecation)

Validation suggestion: run 10–20 historical site analyses (with known scaling or no-scaling outcomes) through the corrected engine and through an established membrane projection program or PHREEQC, and publish the comparison internally before Flux presents the results as projections to customers.

Line numbers refer to commit 030a52a. The benchmark script (docs/review/benchmark_cases.py) now runs the v2.0.0 engine; v1 results are kept in Section 2 for reference.